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Efficient Gait Recognition with SEGait-ConViT: A Vision Transformer Model Enhanced by Sparse Edge Information

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Abstract

Accurate identification of a person from huddled scenarios like surveillance camera feeds is one of the arduous techniques in the domain of computer vision. The gait of a person is a reliable biometric recognition technique, enabled by its capability to perceive a target from a few meters away. Gait analysis can recognize a person from 5 to 10 meters away, depending on environmental factors. Vision Transformers, an alternative approach that can overcome the shortcomings of convolutional neural networks, mandate time-intensive, pricey pre-training on massive datasets. This research proposes a Vision Transformer model for gait-based human recognition. The proposed Vision Transformer model uses gait energy images as input, combined with edge information and sparse regions within those edges. The SEGait-ConViT integrates random convolutions into the self-attention layers and leverages optimized hyperparameter values to improve performance. Gait3D and CASIA-B, two popular public datasets, were utilized to assess the model's efficacy. The suggested methodology achieves state-of-the-art achievement, as per the empirical results which is 66.5% accuracy on Gait3D and 98.8% accuracy on CASIA-B dataset.

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Keywords: CNN; Convolutional Vision Transformer; Gait Recognition; Gait Energy Image; Deep Learning;

1. Introduction

Face identification is one of the predominant techniques employed for the identification of people by their biometric parameters. The human face is perhaps the most familiar and convenient biometric feature in preference to fingerprints or retinal scans in this age of digital photography. Even while facial recognition software is widely used in many

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industries, it has a number of serious drawbacks that reduce its usefulness and provide serious moral dilemmas. Because these systems present serious data security problems and can be used for mass surveillance without people's consent, privacy concerns are crucial. Furthermore, certain demographic groups experience higher mistake rates due to intrinsic biases in training datasets, which promote prejudice and exacerbate social inequality. Environmental elements, including changes in illumination, camera angles, and occlusions, can impair performance, leaving these systems open to misidentifications, particularly when people wear masks or sunglasses. Furthermore, spoofing and adversarial attempts, such as using images, videos, or 3D masks to trick the computers, might affect facial recognition technologies. Their scalability and real-time applicability are limited by their high computational and resource requirements, especially in contexts with limited resources. Additionally, these systems have trouble generalizing because models that were trained on certain datasets would not be able to adjust adequately to a variety of populations or evolving individual appearances over time. Another level of complexity is introduced by the ethical and legal ramifications of consent, autonomy, and adherence to data protection laws. Finally, the issues with reliability, such as false positives and negatives, erode confidence in these technologies and draw attention to the possible repercussions of errors. All of these drawbacks highlight the necessity of implementing facial recognition technology with caution and responsibility, placing a strong emphasis on security, justice, and individual rights.

Other biometric techniques such as Iris and Fingerprint are also vulnerable to attacks. In the case of Iris recognition, an attacker can pretend to be another person using his own iris with zero effort, or use photo, video, or fabricated iris [1,2]. In Fingerprint-based recognition methods, artificial fingerprints can fool the fingerprint recognition machine easily [3,4]. A person's gait is unique, not easily replicable by other people; hence it is an optimistic method for personal identification. Unlike traditional biometrics, gait recognition does not require physical contact or any type of body sensors to recognize a person, if we follow silhouette-based gait recognition technique. When people are reluctant to give their facial identification or are reluctant to employ biometric techniques that require touch or direct skin contact, gait recognition turns out to be a useful substitute. Furthermore, gait recognition provides a continuous real time recognition that obviates any specific action or interactions, like fingerprint and iris scanning. Gait recognition can be used on grainy or low-resolution videos, as it needs only the walking style of the person. Besides, silhouette-based gait recognition can be easily integrated with existing surveillance systems, devoid of any extra sensors or hardware. Vested by these properties, gait recognition is an easily deployable, cost-effective biometric technique. In addition to recognition of a person, gait can be manipulated to deduce diverse information, such as gender identification, walking style detection, and health condition [5]. Earlier research on human-gaits were limited to analysis of the gait and identification of any disorder, compared to normal adults and elderly people. [6]. Gait recognition can be accomplished in three different ways [7]:

1.1 Temporal alignment based

One of the important terms used in person gait recognition is gait cycle. The cycle of the gait starts when the foot hits the floor and ends when the foot strikes the floor again. Temporal alignment-based methods consider a gait cycle as a time series. The features considered for this are different forms of silhouettes, such as principal components of silhouette boundary vectors, width vectors of silhouettes, and Fourier descriptors of the silhouette. These sequential features, since they have a specific order, will be aligned using different techniques such as simple temporal correlation, dynamic time wrapping, Hidden Markov Model, or Fourier analysis. The distance between two sequences is then calculated using these aligned characteristics. Several distance measures, including the dot product and Euclidean distance, are used to calculate the distance between the sequences.

1.2 Silhouette or gait energy image (GEIs) Based

A human silhouette is a simplified representation of a person's form, which is focused on the outline and shape of the person's body rather than detailed features. The silhouette shape-based approach involves considering the silhouettes of a gait cycle and generating the average image, which is known as GEI. The similarity between two GEIs can be calculated by finding the Euclidean distance, linear discriminant-based distance, or using deep learning (DL) classifiers [8].

1.3 Static Parameter Based

The static parameter-based approach considers static parameters such as stride speed, length of stride, and cadence. Other static parameters include different body part size ratios, such as shoulder-to-hip ratio, upper arm-forearm ratio, head-body ratio, upper leg-to-lower leg ratio etc. Since this approach needs information from 3D calibration, it will not give competitive accuracy on common datasets. Gait recognition using deep neural networks is an active area of research, using convolutional neural networks (CNNs). CNN [9] is the traditional DL model, which is principally used for analysis of images and videos.

CNN can be used for classification [10] and segmentation [11] of images, detection [12] and recognition of objects. CNNs have hard inductive bias, such as locality, weight sharing, translation invariance, and translation equivariance, which will increase their capability to be sample efficient. Despite yielding better results on diverse vision-based tasks, CNN has limitations as well. CNN has limited receptive fields and a lower performance ceiling. Generally, CNNs process input images in an ordered way, which compromises the resolution gradually in the pooling layers. For high-resolution images, obligate computation-intensive premiums.

Visual Transformers (ViTs) [13] are another type of DL model that is contributing substantial results in the field of computer vision. ViTs are extended versions of transformers, which have afforded significant contributions to natural language processing (NLP). Marshaling soft self-attention layers, ViT models outperform CNNs by four times, in accuracy and computational efficiency [14]. Nevertheless, ViTs call for large-scale data sets to achieve remarkable results, yet they can be computationally expensive as they solicit self-attention mechanisms. Perceptively, the integration of CNNs and ViTs can leverage local and global feature-capturing capabilities—CNNs are able to capture local patterns such as shape, texture, and edges of images, whereas ViTs are empowered to capture global features like understanding the semantics and overall context of the image. Combining these two is expected to improve overall performance by enabling the model to extract local information and global information from input image. The combination of CNN and ViT is termed the Convolutional Vision Transformer (ConViT) [15]. ConViT possesses positional self-attention which has gating mechanism, which is basically a soft convolutional inductive bias type of self-attention. We are proposing a model named ‘Sparse Edge Gait Convolutional Vision Transformer’ (SEGait-ConViT), which is highly inspired by the ConViT model with substantial modification for accurate gait recognition. The proposed SEGait-ConViT was evaluated on two public gait data sets: CASIA-B, the first one, originated in 2005 and is a multi-view gait recognition dataset, consisting of 124 silhouettes of objects with different variations—angle, carried object, and clothing. Silhouettes of a person are binary images of the person that furnish the shape of a 2D image. OU-ISIR, the second one, is another gait dataset developed by Osaka University, composed of 5,114 males and 5,193 females from different age groups, having 14 view angles.

We have used GEIs of person as the input. GEI is one of the gait features, which comprises constant and active information of a gait sequence. Static information includes the shape of a body, while the active or dynamic information comprises changes in the frequency and phase. GEIs are aggregated, time-normalized energy images of complete walking cycles. Prime contributions of the paper are noted hereunder:

- Proposed an architecture named SEGait-ConViT for gait-based person identification. This architecture is a modified version of the Convolutional Vision Transformer. In SEGait-ConViT, we have initialized the self-attention layers using random convolutions. This is the first gait-based person recognition system that we are aware of that uses sparse edges as an input along with the normal GEI image data to the modified ConViT.
- Tested the accuracy of person recognition using SEGait-ConViT with gait datasets such as Gait3D and the CASIA Gait dataset-B. Also balanced the Gait3D dataset using augmentation techniques.
- Tuned the hyperparameters of the SEGait-ConViT for better accuracy. The contents of this paper are organized into several sections.

The ’Related Work’ section encompasses the prior research conducted by different authors that relates to gait recognition. Subsequently, the ’Methodology’ section will provide an in-depth explanation of the proposed SEGait-

ConViT architecture. The 'Experimental Results' section will outline the outcomes obtained from applying SEGait-ConViT to the CASIA Gait dataset-B and Gait3D gait datasets. Finally, the 'Conclusion and Future Work' section will elucidate the discoveries made in our study and lay out potential directions for future endeavors.

2. Related Work

Gait related work can be classified from 3 aspects: feature, DL, and attention mechanism. Sourabh A. Niyogi et al., developed one of the earlier feature-based gait recognition systems in 1994. Sourabh's team modelled a person using the local structure of spatio-temporal pattern [16]. The feature-based techniques can be categorised mainly into model based and appearance based. In the first method (model), mathematical models are used to identify a person using the walking style. Modelling involves feature extraction, which involves extracting a person's waking pattern using different parameters such as motion of body parts, trajectory of limbs, angles of joint movement, cadence, stride duration, step length etc. Using the extracted features, a mathematical model will be generated. This model encodes the unique patterns and parameters of a person's walking style. R. Tanawongsuwan et al. did model based methods by manually marking different body parts [17]. They did classification using Nearest Neighbour using Euclidean distance measure. Wang et al. [18] built a model from 3D coordinates of the human skeleton, they have included 21 joints and on a private dataset they have achieved 92.3% accuracy using Nearest Neighbour classification method. In [19] Zhen et al. utilized the stride width, angle and elevation of joints etc. as the static and dynamic information and they used deterministic learning algorithms for classification. On the other hand, a frame level matching was presented by Choi et al. [20]. Gaitset [21] used gait data as a deep set; which is a collection of images, for person recognition. They have achieved 96.1% on CASIA-B and 87.9% on OU-MVLP. In [22] Rida et al. has used the Globality-Locality Preserving Projections (GLPP) algorithm. They got 86.06% accuracy on CASIA dataset B. In [23] Zhang et al. proved that a model generated using selective spatial-temporal features gives better performance than GEI. They have used long short-term memory for classification. In [24] Chen et al. designed a method to recognise the gait of multiple people walking. They have used the support vector machine framework for the classification purpose.

In [25] Zhu et al. did a work on enhancing the appearance-based gait recognition technique named GaitHF used human height data for recognition purpose. In Deep learning-based methods, the models are capable of learning features directly from raw data without specific feature engineering. Shiraga et al. used one of the popular deep learning methods CNN to recognise gait of a person [26]. They have used GEI as input to the model and verified it using the OU-ISIR dataset. In [27] Wolf et al. developed a gait recognition method using 3D CNN, which is invariant to the person's speed, walking, clothing and view. They have tested the model on CASIA-B dataset and got 94.3% to 99.9% accuracy on different viewing angles. In [28] Ding et al. proposed a sequential convolutional network on video data for gait recognition. It achieved 83.8% accuracy on the OU-MVLP dataset and 95.2% on CASIA-B dataset. In [29] Mogan et al. did a pretrained VGG-16 model based gait recognition using multi-layer perceptron. It got 99.1% accuracy on the OU-LP dataset. In [30] Elharrouss et al. proposed a multi-task CNN for person re-identification using GEIs. They have achieved 99% accuracy on CASIA-B, and in OU-ISIR and OU-MVLP dataset, they achieved 98% accuracy.

Traditionally, Attention based Models were used in NLP based research, which are basically transformers. Recently it has got 3 ten good acceptances in image processing. Wang et al. proposed a self-attention mechanism for gait recognition [31]. They have used GEIs for the same. The method got an accuracy of 91.7% on CASIA-B dataset. Vision transformer is a transformer which is made for computer vision-based tasks which is called ViT [32]. ViT splits an image into several pieces/patches, which then serializes each piece into a vector, and then translates it to a lesser dimension vector using a single matrix multiplication. In terms of accuracy and pre-training cost, ViT better than SOTA convolution-based architectures (ResNet) in image classification tasks. In [33] Prasetyo et al. examined whether sparse regularization can be applied to vision transformers and how pruning affects the efficiency vs. performance trade-off, either before or after sparse regularization. Hatamizadeh, Ali, et al. presented FasterViT [34] a hybrid vision transformer that aims to achieve the best possible balance between throughput and performance. Their

hierarchical Attention module, has the capability to extract long-range spatial dependencies and cross-window interactions of local regions.

Another transformer that combines the benefits of CNNs with Transformers is explained in CvT [14]. Transformers are having good generalization capacity, dynamic attention mechanism and capacity to fuse the global context. Whereas the CNNs are good spatial sub sampling, weight sharing and local receptive field. CSWin Transformer [35] is another Transformer-based backbone for vision applications. By using "Cross-Shaped Window self-attention," it analyses the entire image at once. DeiT [36] is a data-efficient training process for image transformers that eliminates the need for a substantial amount of training data. DeiT is an approach that aims to create a convolution-free model that can outperform convolution-based networks while requiring less training data. Heo, Byeongho, et al. introduced PiT (Pooling-based Vision Transformer) [37] which will increase processing performance by gradually reducing the number of tokens by adding pooling layers to the transformer. By gradually aggregating tokens to represent images at various scales, T2T-ViT [38] aims to improve the tokenization process.

3. Proposed methodology

CNNs are largely used for image classification tasks, because of the convolution's innate power of feature extraction. Convolutions refer to transformation of an image by mathematical operations using a kernel; kernel or filter is a small matrix, which will slide over the input image matrix, multiply them element wise, and transform the image. Google Research Brain [13] put forward an idea that transformers, ViTs can, not only be used for text classification, but for images and videos as well. ViT achieved remarkable results for classification tasks in computer vision. ConViT is a hybrid model of ViT and CNN. This paper proffers the idea of gait-based person recognition, using SEGait-ConViT, the modified and optimized layers of ConViT. The recommended methodology employs SEGait-ConViT to train the GEI which is generated using silhouette data. Gait-based person recognition using ConViT involves diverse steps, such as pre-processing, model selection, optimizing the hyperparameters, forward pass, and result analysis. In this scenario, the input data is a GEI type binary image. DL models like ConViT are primarily set up to handle colour/RGB images. With the help of OpenCV, the input binary GEI images will be converted to a three-channel, RGB format. This RGB conversion is for utilizing the pre-trained weights effectively. Also, it will maintain the model architecture without modification. ConViT consists of three types of pre-trained models: convit_tiny, convit_small, and convit_base, which have 4, 9, and 16 heads, respectively. This paper used convit_base for prediction.

Rationale for Choosing convit_base Over convit_tiny and convit_small:

- Enhanced Feature Extraction Capability: Compared to convit_tiny and convit_small, convit_base has larger number of parameters and deeper layers which will increase the capacity to learn more complex features. These additional layers are critical for capturing both local and global gait dynamics.
- Superior Performance Metrics: Benchmark evaluation in [15] demonstrates that convit_base outperforms its smaller counterparts.
- Scalability for Multi-Class Recognition: it can handle multiclass classification which will be normal in real-world applications (e.g., security, surveillance).
- Alignment with SEGait-ConViT's Objectives: Convit_base's robust feature extraction capabilities compliment the SEGait-ConViT architecture, ensuring that important gait patterns are extracted and fed accurately into the transformer layers for further processing. The depth and sophistication of convit_base support the implementation of complex self-attention mechanisms of the proposed model.

After the model selection, input images are passed to this model and will do the operations using attention mechanisms and convolutional layers to extract important features. After the extraction of relevant features, the model will predict class-based on the learned features. After getting the predicted model, the next stage is to apply the SoftMax to get the class probability for the input image.

3.1. Pre-processing of data

Silhouette preserves the appearance of the body, which plays an important role in gait identification process. The silhouette is generated from the video data. Including silhouette of people with backpack, different clothing condition, male and female etc will increase the diversity of the dataset. GEI is the average all the silhouette images in a gait cycle of each person. It preserves the complete information of a gait sequence, including static information such as physical aspects like shape of the human body, and dynamic details such as phase and frequency of gait sequence.

Following steps are part of the data pre-processing steps: Create GEIs from a series of silhouette images that contains a person's complete gait cycle, enhancing model robustness by implementing both standard and dataset-specific augmentations, resizing the images into 224x224 and normalization. This method uses min-max normalization, which is the best normalizing strategy for binary GEIs. It works well with deep learning models scaling to [0, 1] and preserves binary information.

- CASIA-B Augmentation Strategies: CASIA-B data are having a controlled environment which has consistent backgrounds and lighting and it has specific walking styles such as casual walking, put on a coat and having/carrying a bag. CASIA-B needs Condition-Specific Augmentations such as simulate carrying a bag, simulate wearing a coat and geometric transformations.
- Gait3d Augmentation Strategies: Gait3D is captured in unconstrained, real-world environments with diverse backgrounds, lighting conditions, camera angles, and potential occlusions. So, the augmentation needed for this dataset are: 1. Geometric Transformations: Random rotation with range: ±20° to ±30° which will reflect the wide range of camera angles and walking directions in real-world scenarios. 2. Random Scaling: Range: 90% to 110% of original size which will simulate variations in subject distance from the camera. 3. Random Translation: Range: ±10% of image dimensions horizontally and vertically, which will mimic diverse subject positions within the frame.4. Random Horizontal and Vertical Flip.

3.2. ConViT

ConViT, an amalgam of ViT and CNN, this hybrid model is used in different computer vision tasks. Inductive bias and self-attention are the terms considered in ConViT to improve its performance. Inductive bias: is a set of definite or indefinite assumptions made in the algorithm which help in generalization of the model. There are basically two types of inductive biases: hard inductive bias and soft inductive bias. Hard inductive biases are the explicit assumptions imposed on the model. If the data set to be analysed is diverse in nature, like it's having subgroups, classes or categories with different characteristics; then hard inductive bias will be constraining. Because hard inductive biases always look for a specific aspect, which will miss out on a wide variety of patterns of the data. In such cases it is better to choose soft inductive bias, because soft inductive bias can adapt to varying patterns which are present in the diverse dataset. ViT has hard inductive bias whereas ConViT has a soft inductive bias, which promotes robustness of ConViT.

Self-attention (SA) is a technique in transformers for extraction of relationships within a sequence. ConViT comprises gated positional self-attention (GPSA), an extended self-attention module, which has a convolution-based soft inductive bias. Generally, elements in self-attention are connected through the relationships between them, instead of their positional information. GPSA holds positional information along with self-attention mechanism. A gating parameter is present in the GPSA layers, which are used to regulate the position and content of an image. The ConViT [15] is a ViT variant, in which GPSA layers replace SA layers, which will be in any ratio. This architecture comprises 10 GPSA layers and 2 SA layers. As positional information is innate with GPSA layers, Class tokens, specific tokens which furnish positional information, are provided for SA layers only. The input silhouette image size is selected as 224 x 224 (width X height). ConViT slices the image into patches of size 16 x 16, followed by embedding these images into vectors The dimension of these image patches is 64Nh. ('N' is the # of patches into which the image is divided. 'h' represents the number of attention heads). These positional embedding vectors form inputs to the transformer encoder. Pursuant to encoding, a multilayer perceptron will perform the classification task. Multi-layer perceptron's are a fundamental architecture in DL and are used for different tasks such as classification, regression

etc.

*3.3. Sparse Edge Gait Convolutional Vision Transformer (*SEGait-ConViT)

This paper introduces SEGait-ConVi, which is a variant of ConViT designed to recognise gait of a person. Initialize the SA layers with random GEIs + convolutions. Random convolutions increase the diversity of initial weights, and hence it will make the model more reliable. It will ensure that the initial weights of SA layers are unbiased towards specific patterns/features. Also, it will make the model fit for extracting wide range spatial relationships and patterns. Adding random convolution to the SA layer is basically a type of noise injection to the network. This will act as a regularization and will help in preventing overfitting of the model.

The architectural flow diagram of SEGait-ConViT is depicted in Fig. 1 and feature mining diagram is shown in Fig. 2. It is a modified version of ConViT. The model has 16 X 16 patch sizes with 10 Gated Positional Self-Attention layers and 8 numbers of heads. The SA layer is the fusion of random convolutional self-attention and traditional self-attention. The depth of the Vision Transformer is 12 and the multi-layer perceptron ratio is 4. Input sequences are encoded by embedding layers into a sequence of dense vectors. The dimension of embedded data is 48. And the comprehensive layer-wise input-output dimension of the proposed architecture for Gait3D dataset is shown in Table 1.

Table 1. Comprehensive Layer-Wise Input-Output Dimensions on Gait3D dataset.

Layer Name	Layer Type	Input Dimension	Output Dimension
1.Input Layer	Input	64×64×1	64×64×1
2.Binarization	Adaptive thresholding	64×64×1	64×64×1
3.Blurring using Gaussian technique	Gaussian Blur	64×64×1	64×64×1
4. Edge detection using Canny Edge algorithm	Canny Edge Detector	64×64×1	64×64×1
5. Dilation	Dilation	64×64×1	64×64×1
6. Sparse Edge Computation	Subtraction	64×64×1	64×64×1
7. Combine Edges	Maximum Operation	64×64×1	64×64×1
8. Patch Embedding	Conv2D	64×64×1	4×4×768
9. Flatten Patches	Flatten	4×4×768	768×16
10. Transpose for Sequence	Transpose	768×16	16x768
11. CLS Token Preparation	Parameter (Learnable)	-	1x768
12. Concatenate CLS Token	Concatenation	16x768	17x768
13. Position Embedding Addition	Position Embedding	16×768	17x768
14. Dropout	dropout	17x768	17x768
15. Random Convolution in Attention	Random Convolution Layer(x12)	17x768	17x768
16. Transformer Encoder Blocks	Transformer Encoder Blocks x12	17x768	17x768
17. Layer Normalization	LayerNorm	17x768	17x768
18. Classification Head	Linear (Fully Connected)	768	3000
19. Softmax Activation	Softmax	3000	3000

Self-attention layer with random convolution is an uncustomary approach which has improved the accuracy of our model. The architecture of the proposed methodology has different steps. They are:

- Input data: The input will be the GEI image and the sparse edge of the GEI image, in order to get the movement information of the person. *Edge and Sparse Detection:* Given a binary input GEI image where each pixel value will be either 0 or 255. Convert GEI image to a NumPy array if it is not already in the form. Ensure the image is binary form. To reduce the noise and detail of an image, apply gaussian smoothing. The

gaussian blur is applied via convolution with gaussian kernel K and standard deviation σ. Apply the Canny edge detector to the blurred image GB. After the blurring of image, compute the gradient magnitude and direction. To distinguish between strong and weak edges, use thresholding. If weak edges are linked to strong edges using hysteresis edge tracking, they are kept in place. *Sparse Edge Detection:* Use dilation and subtraction to find sparse areas along the edges. Generate an image by combining by detected edges and sparse region. The whole edge sparse feature extraction is shown in Fig. 3. It shows the GEI of the person, detected edge and the sparse area of the same. *Gaussian Blur Parameters:* The kernel size used is 5x5 and σ = 1.0 which offers a balanced approach between noise reduction and edge preservation. *Canny Edge Detection Parameters:* Canny Edge Detection is employed to identify and highlight edges within the Gaussian Blurred GEI. The parameters here used are Low Threshold Ratio (T_L) and High Threshold Ratio (T_H). T_L is the lower bound for detecting edges. Pixels with gradient magnitudes above this value are considered as edge candidates and the selected value is T_L = 0.10. T_H is the upper bound for detecting strong edges. Pixels with gradient magnitudes above this value are classified as strong edges and the selected value is T_H = 0.30. The full feature extraction process of the model using sparse edge is shown in the Fig. 4 (Person having coat), Fig. 5 (Person carrying bag) and Fig. 6 (Person Normal Walking). In each of these figures, first sub-image is the original GRI, then Gaussian blurred GEI, Canny edge detected image, detected edge image, sparse edge image and finally the combine edge sparse image are numbered from a to f.

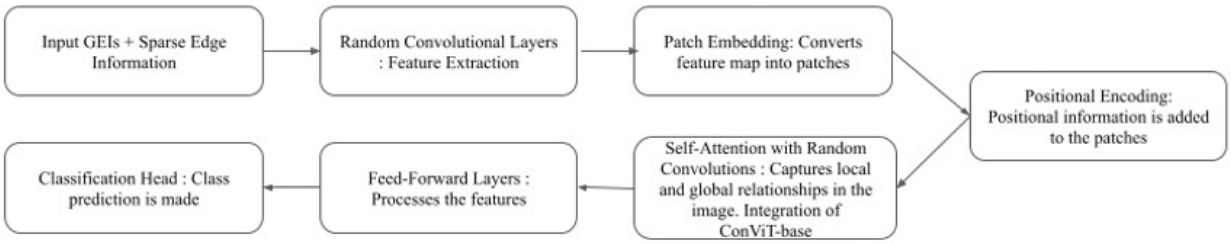


Fig. 1. SEGait-ConViT Framework Overview.



Fig. 2. SEGait-ConViT feature extraction Overview.

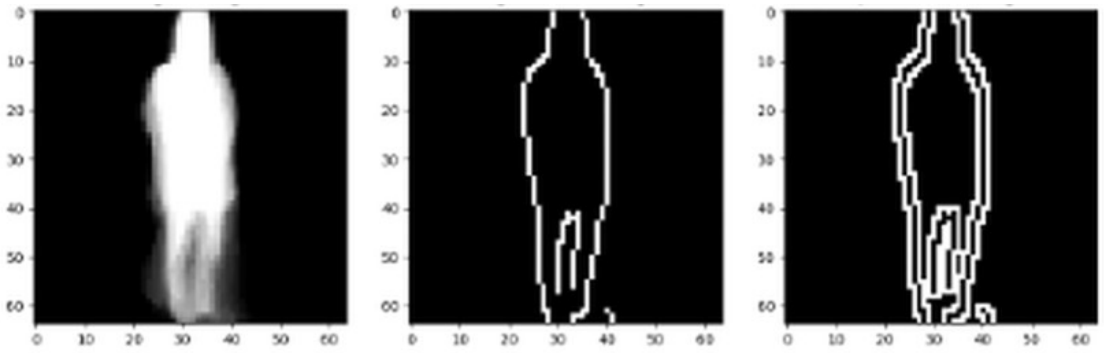


Fig. 3. Edge sparse feature extraction: GEI, Edge and Sparse area.

- Random Convolutional Self-Attention: Split the input data sequence into patches of size 'S'. Then we will apply random filters for each patch to get convolutional outputs:$Conv_{out} = [rc_1, rc_2, rc_3, \ldots\ldots, rc_N]$ where each rc_i is the output of random convolutions for the i-th patch. The random convolutions will be able to capture different local patterns of the data, because of its property of randomness. After applying random convolutions, the attention score of each patch will be calculated using dot product between queries and keys. Attention score Compute query, key, and value projections for the convolutional outputs. Query projection, $Q_{rc} = rc_i * Q_{w\,rc}$, key projection, $K_{rc} = rc_i * K_{w\,rc}$, value projection, $V_{rc} = rc_i * V_{w_rc}$ Compute self-attention scores for each patches using dot products.

  Attention matrix, $Att_{score} = softmax(Q_{rc} * K_{rc}^T)$ (I)

  Then calculate weighted sum of values. Weighted sum of values, $O_{rc} = Att_{score} * V_{rc}$ $Q_{w\,rc}$, $K_{w\,rc}$, $V_{w\,rc}$ are weight matrices for query, key, and value projections of random convolution, which are learnable. After calculating random convolutional self-attention, calculate traditional Self-Attention: Traditional Self-Attention: Compute query, key, and value projections for the entire input sequence.

  query projection, $QU_p = \mathrm{X} * Q_{w_p}$, key projection, $KE_p = \mathrm{X} * K_{w_p}$, value projection, $VA_p = \mathrm{X} * V_{w_p}$

  Compute attention scores $Att_t = softmax(QU_p * KE_p^P)$ (II)

  Compute the weighted sum of values: $OU_v = Att_t * VA_v$; $Q_{w\,v}, K_{w\,v}, V_{w\,v}$ are weight matrices for query, key, and value projections of traditional convolution methods, which are learnable.
- Amalgamation with traditional self-attention: We uses a gating function to combine the outputs of random convolutional self-attention (O_{rc}) and traditional self-attention (O_t). These traditional self-attention layers are able to capture global relationships and dependencies between tokens. Calculate gating factor using sigmoid activation function, $G_f = sigmoid(W_g * [O_{rc}; O_t] + b_g)$, where W_g and b_g are learnable weights and biases of gating units.

  Compute amalgamated output $Am_{output} = G_f * O_{rc} + (1 - G_f) * O_t$ (III)

- Feed-forward Neural Networks: After calculating the combined output of random convolutional self-attention and traditional self-attention, it will be passed to feed forward neural network, to process the information: $FFN_{output} = FFN(Fused_{output})$
- Output Layer: In the output layer we will apply a multi-layer perceptron for the classification task.

  $Output_{predictions} = Output_{Layer}(FFN_{output})$ (IV)

3.4. SEGait-ConViT handles occlusions

- GEIs as Input: Temporal Integration is done by combining gait data from several frames, GEIs can efficiently eliminate transient occlusions. Because of this temporal averaging, the overall depiction of gait is not disproportionately distorted by sporadic occlusions. Consistent Feature Representation is done by GEIs reduce the variability caused by sporadic occlusions by compressing motion into a single image, which offers a steady input for feature extraction.
- Sparse Edge Feature Extraction (SEG): It will focus on Prominent Features by model using sparse edges to highlight important gait structural elements including body position and limb movements. This concentration aids in preserving the integrity of crucial gait traits even in situations where portions of the silhouette are momentarily obscured. Noise Reduction: The model's capacity to focus on significant gait patterns is improved by sparse edge extraction, which reduces the impact of extraneous or noisy background features.
- Occlusion-Inclusive Data Augmentation: To increase the generalization capacity of the model by increasing occluded circumstances during training, data augmentation strategies introduce different occlusion scenarios (e.g., masking parts of the GEIs). The model learns to identify and adjust to partially visible or missing gait cues as a result of this exposure. SEGait-ConViT maintains high accuracy under moderate occlusion levels and is more resilient to similar real-world challenges by learning from augmented data that includes occlusions. Fig. 7 provides an example of occlusion on the Gait3D dataset. Random rectangular occlusion is the occlusion technique employed in Fig 7.

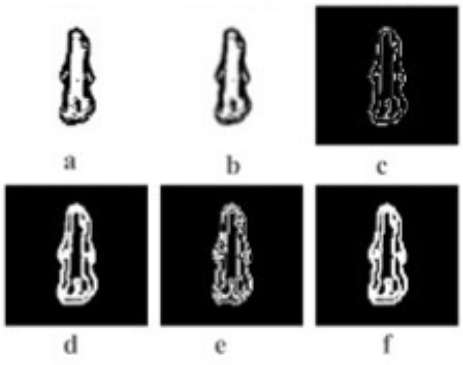


Fig. 4. Sparse edge person having coat.

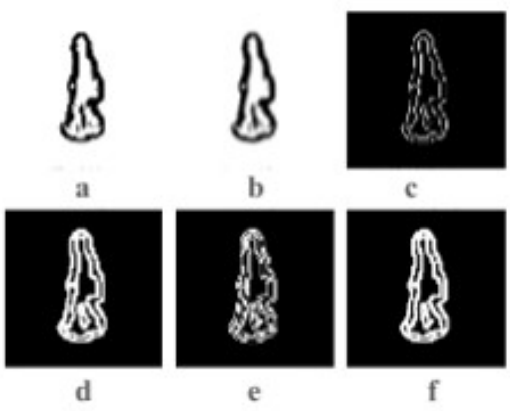


Fig. 5. Sparse edge person carrying bag.

(a)Original gei b) Gaussian blur c) Canny edge d) Dilated edge e) Sparse edge f) Combined edge sparse

3.5. Feature extraction and hyperparameter tuning in SEGait-ConViT

Sparse edge gait is used in SEGait-ConViT's feature extraction procedure to efficiently collect crucial motion patterns needed for precise gait detection. GEIs, which show the temporal average of a person's walk across several frames, are used to extract sparse edge features. By emphasizing sparse edges, the technique reduces the impact of duplicate information and background noise while highlighting important structural and dynamic gait features like torso rotations, stride patterns, and limb movements. This selective focus enabling the algorithm to discern minute

differences in walking patterns between people.

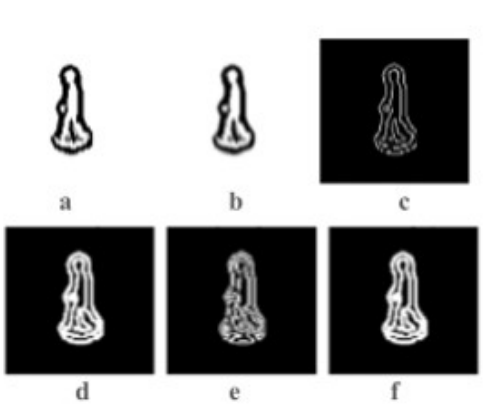


Fig.6. Sparse edge person with normal walk.



Fig. 7. Original image and after occlusion.

Fig. 6. a) Original gei b) Gaussian blur c) Canny edge d) Dilated edge e) Sparse edge f) Combined edge sparse.

4. Experimental results

This section describes the data sets which were used for training SEGait-ConViT and tuning the hyper-parameters for evaluation of the performance of it. The experiments were conducted on Microsoft Visual Studio Code. A single node machine with 4 NVIDIA Tesla V100 Volta GPU Accelerator 32GB Tesla nnV100-PCIE-32GB0 Graphics Card was used. The NVidia Driver version was 515.43.04 and CUDA version 11.7.

4.1. Datasets

The open-source CASIA-B Gait dataset was used for evaluation of the accuracy of the proposed SEGait-ConViT method. Which consists of 124 persons captured in 11 different views. The variations considered were view angle, with coat, and carrying bag. Each person was curated with 8000 images, a total of 992,000 (= 8000 x 124) images were selected. In this the train and test proportions were selected as 80% and 20% respectively. Gait3D [39] is a large-scale dataset that includes over 25,000 sequences and 4,000 subjects that were recorded by 39 cameras in an unrestricted indoor setting. The Gait3D split into a training and test dataset, having the counts of 3000 and 1000 respectively. One sequence per subject is chosen at random to produce a probe set of 1,000 sequences for testing, with the remaining sequences being assigned to the gallery set. To increase the agility and generality of SEGait-ConViT, pre-processing gait datasets entails standardizing the data format and implementing augmentations. Gait3D and CASIA-B have separate pre-processing stages since Gait3D contains real-world scenarios and CASIA-B has controlled ones.

4.2. *Tuning hyper parameters*

The hyper-parameters which were tuned in the model were input image size (I), batch size (b), optimizers (θ), learning rate (lr) and weight decay (w). Grid search technique was used to select ideal/optimized hyperparameters. Grid search technique is the technique where the value of only a single hyper-parameter is changed, while those of other selected hyper-parameters are kept constant. Accuracies of the SEGait-ConViT model, at different batch values, are depicted in the first and second columns of the Table 1. Batch size 32 gave the highest accuracy. Increased batch sizes entail high computational cost, poor generalization, and compromised accuracy. Third and fourth columns of Table 1 show the accuracy of the SEGait-ConViT methodology with different optimizers θ. Adamw performed better

than Nadam and Adam optimizers. Adam with decoupled weight decay and loss based gradient update is Adamw. Accuracies of the methodology with different learning rates (lr) are shown in the fifth and sixth columns of Table 1. It shows how to change the proposed model with respect to the error value in every weight update. When the smallest value of 0.00001 was selected for the model's learning rate, longer training times were observed. In contrast, when selection of the highest value of 0.01, for lr, resulted in learning a sub-optimal weight. The learning rate value 0.001 gave the maximum performance. Weight decay value is undertaken to avert over-fitting, by reducing the complexity of the model. The different weight decay (w) values were 0.0001,0.001,0.01 and 0.1. The results are shown in the Table 2; column number 7 and 8.

The optimized value of weight decay in the model is 0. 0001.The columns 9 and 10 of the Table 1 show characteristics of the accuracy of the proposed SEGait-ConViT at different values of input sizes. The tested input sizes were 64 x 64, 128 x 28, and 224 x 224, with the ideal value turning out as 224 x 224. The finalized hyper parameter values were chosen by considering the accuracy and running time of the model. And the optimal values are shown in Table 3.

Table 2. Tuned Hyper Parameter Value and accuracies for Batch Size, Optimizers, Learning rate, Weight Decay and Input size on CASIA-B.

Batch Size	Accuracy (%) I = 224 X 224, lr = 0.001, θ = Adamw, w=0.0001]	Optimizer	Accuracy (%) [I = 224 X 224, lr = 0.001, θ = Adamw, w=0.0001, b=32]	Learning rate	Accuracy (%) [I = 224 X 224, θ = Adamw, w=0.0001, b=32]	weight Decay	Accuracy (%) [I = 224 X 224, θ = Adamw, lr=0.001, b=32]	Input Size	Accuracy (%) [θ = Adamw, lr=0.001, w=0.0001 b=32]
32	98.95	Nadam	98.43	0.00001	98.43	0.0001	98.95	64X64	98.43
64	89.52	Adam	98.43	0.0001	98.52	0.001	98.43	128X128	98.52
128	98.43	Adamw	98.95	0.001	98.95	0.01	98.52	224 X 224	98.95

Table 3. Tuned Hyper Parameter Values.

Hyper Parameters	Tested Values	Ideal Values
Input Size(I)	64X64,128X28, 224 X 224	224 X 224
Batch Size(b)	32,64,128	32
Optimizer(θ)	SGD, Adam, Adamw	Adamw
Learning Rate(lr)	0.00001, 0.0001,0.0005,0.001, 0.01	0.001
weight decay(w)	0.0001,0.001,0.01,0.1	0.0001

4.3. *Accuracy*

Attainment of 100% accuracy is generally impossible for any DL or transformer model, in real world scenarios. We have considered Top-1 and Top-5 accuracy models. Fig. 8 and Fig. 9 show the accuracy and loss graphs of the CASIA_B dataset. Rank-1 accuracy is the proportion of queries where the correct identity appears at the top position in the list of retrieved results. In the case of person identification, it calculates how often the model correctly identifies a person as the most similar match from the gallery of images.

Rank-1Accuracy= (Number of correct top-1 matches/ Total number of queries) ×100%

Reasons for Prioritizing Rank-1 Accuracy:

1. Direct Measure of Correct Identification: This simple technique assesses the model's capacity to retrieve the top candidate that was correctly recognized.
2. Operational Relevance: Finding the right person on the first attempt is frequently the most important criterion in real-world applications, particularly surveillance.
3. Simplicity and Interpretability: It is easy to understand and communicate, making it a good metric in many research studies and deployments.
4. Benchmarking Standard: It is a widely used person REID benchmark.

Fig. 10. Shows the confusion matrix for Top 10 classes of CASIA-B dataset.

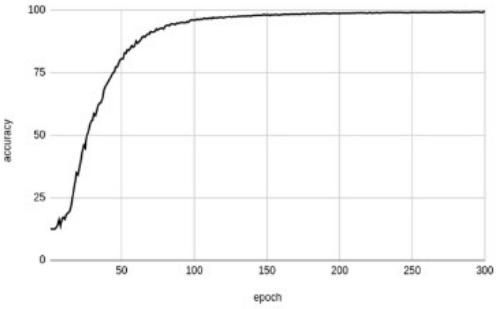


Fig. 8. Accuracy graph of CASIA-B dataset.

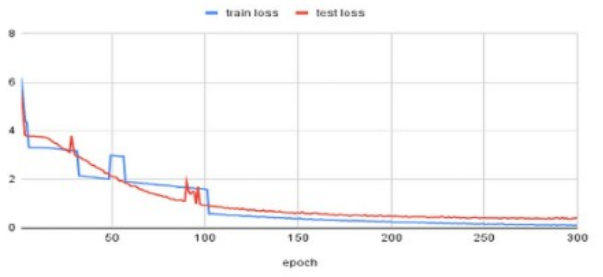
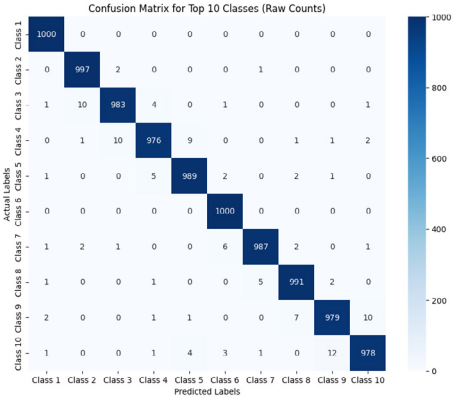


Fig. 9. Epoch-Loss graph of CASIA-B dataset.



. Fig. 10. Confusion matrix for Top 10 classes CASIA-B dataset.



Fig. 11. Extreme occlusions from Gait3D dataset.

4.4. *Comparison with existing method*

The experiment was conducted on the Gait3D dataset, and our method provides more accurate information on the dynamic attributes of the person, which is explained in Table 4. Consequently, our approach achieves a 0.2% higher Rank-1 accuracy than DyGait [45], outperforming previous gait recognition methods. It also outperforms MTSGait in Rank-1 accuracy by 17.8% and in Rank-5 accuracy by 14.3%. These findings imply that our technique performs better because the dynamic features it captures are more discriminative. Table 4 also shows the number of parameters in million as well as the model size in MB. Our method performs better in lab circumstances as well as on datasets from real-world scenarios. Accuracy of the proposed SEGait-ConViT model was compared against state-of-the-art techniques like GaitSet [21], GaitPart [40], MT3D [41], GaitGL [42], OpenGait [43], MetaGait [44] and DyGait [45] on CASIA-B dataset in Table 5. In Table 5, the methods GaitSet [21], GaitPart [40], MT3D [41], GaitGL [42], OpenGait [43], MetaGait [44], DyGait [45] and SEGait-ConViT(ours) are represented as “Method 1 to 8”. According to experimental findings, the suggested approach has the best average accuracy across all situations (NM, BG, and CL). We also investigate how well the strategies work under various circumstances. It is evident that as the

external environment varies, the recognition accuracy significantly drops e.g.: GaitGL's recognition accuracy in CL, BG and NM for instance, is 83.6%, 94.5% and 97.4%, respectively. Comparable accuracy for MetaGait framework is 98.1%, 95.2% and 86.9%, respectively. Both of the aforementioned approaches are more susceptible to the effects of the complex external environment since they similarly pull information from various parts of the human stride that only contain static data. The percentage of accuracy in normal walk (NM), with bag (BG), and with coat (CL) for the suggested method is 98.4%, 96.2%, and 87.8%, respectively. Another comparable method, the DyGait approach values in BG and CL in terms of recognition accuracy than the other approaches and our method outperformed it. In complex environments, our technique works effectively at certain specified angles (0◦ and 90◦). For instance, our method's accuracy is 94.6% and 94.2% in BG conditions, whereas DyGait's is 94.5% and 94.0%.

4.5. *Methods to further validate SEGait-ConViT*

- Cross-Dataset Evaluation: We can evaluate the model's ability to generalize across different datasets with different distribution, condition and challenges. For example, train on Gait3D and evaluate on other public datasets like OU-MVLP or GREW.
- Fine-Grained Evaluation on Dataset Subsets: Test the model's performance on specific conditions or subsets within a dataset. For example, in CASIA-B Subsets: Evaluate on: Normal Walking Condition (NM): Baseline performance, carrying a Backpack (BG): Ability to handle occlusions. Wearing a Coat (CL): Robustness to appearance changes. Which is shown in Table 5.
- Evaluation on New Gait Datasets: OU-MVLP is a public gait dataset, which includes a wide range of subjects and views. GREW is the real-world surveillance dataset with 26,345 subjects.

Table 4. Accuracy (%), mean average precision, mean inverse negative penalty, number of parameters and model size on Gait 3D.

Techniques	Accuracy (rank_1)	Accuracy (rank_5)	Mean average precision	Mean inverse negative penalty	#parameters in million	Size of model (FP32) in MB
PoseGait [46]	0.24	1.08	0.47	0.34	~15-25	~60-100
GaitGraph [47]	6.25	16.23	5.18	2.42	~10-20	~40-80
GEINet [26]	5.40	14.20	5.06	3.14	~1-5	~4-20
GaitSet [21]	36.70	58.30	30.01	17.30	~25.6	~102.4
GaitPart [40]	28.20	47.60	21.58	12.36	~25.8	~103.2
GLN [48]	31.40	52.90	24.74	13.58	~20-30	~80-120
GaitGL [42]	29.70	48.50	22.29	13.26	~26	~104
CSTL [49]	11.70	19.20	5.59	2.59	~30-40	~120-160
SMPLGait [39]	46.30	64.50	37.16	22.23	~50-60	~200-240
MTSGait [50]	48.70	67.10	37.63	21.92	~40-50	~160-200
OpenGait [43]	65.60	-	-	-	~25.6 - 25.8	~102.4 - 103.2
DyGait [45]	66.30	80.80	56.40	37.30	~27	~108
SEGait-ConViT (Ours)	**66.50**	**81.4**	**60.30**	**40.50**	**~40**	**~160**

4.6. *SEGait-ConViT integration with edge device*

With the right adjustments, it is possible to integrate SEGait-ConViT for real-time gait recognition into edge devices. Through the use of effective frameworks (such as TensorFlow Lite and TensorRT) and methods like quantization and pruning, the proposed model can accomplish: low latency, high throughput, and deployment that uses few resources. Practical uses such as secure authentication, real-time surveillance, and anomaly detection in dynamic contexts are made possible by this integration.

4.7. *Error analysis of SEGait-ConViT*

Extreme occlusion on visible subjects due to environmental objects (e.g., walls, furniture) or other people blocking the view, which will impact the loss of critical gait features (e.g., legs, arms), degrades recognition performance. Fig. 11 displays example images from the Gait3D dataset that have high occlusion and are difficult for the suggested model to recognize. To lessen the impact of the accuracy loss, we have incorporated occlusion into the augmentation in the interim.

Table 5: Accuracy (in %) on various view angles: comparison on CASIA-B (Rank-1) [45], GaitSet [21], GaitPart [40], MT3D [41], GaitGL [42], OpenGait [43], MetaGait [44], DyGait [45] and SEGait-ConViT(ours): Method 1 to 8.

Gallery data NM#1 to 4		0° to 180°											
Probe data		0	18	36	54	72	90	108	126	144	162	180	Mean
Normal walk#: 5 to 6	Method 1	90.8	97.9	99.4	96.9	93.6	91.7	95.0	97.8	98.9	96.8	85.8	95.0
	Method 2	94.1	98.6	99.3	98.5	94.0	92.3	95.9	98.4	99.2	97.8	90.4	96.2
	Method 3	95.7	98.2	99.0	97.5	95.1	93.9	96.1	98.6	99.2	98.2	92.0	96.7
	Method 4	96.0	98.3	99.0	97.9	96.9	95.4	97.0	98.9	99.3	98.8	94.0	97.4
	Method 5	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	97.6
	Method 6	97.3	99.2	99.5	99.1	97.2	95.5	97.6	99.1	99.3	99.1	96.7	98.1
	Method 7	97.4	98.9	99.2	98.3	97.7	96.8	98.2	99.3	99.3	99.2	97.6	98.4
	Method 8	**97.5**	**99.3**	**99.5**	**98.5**	**97.8**	**98.8**	**98.4**	**99.5**	**99.4**	**99.2**	**97.6**	**98.8**
With bag#1-2	Method 1	83.8	91.2	91.8	88.8	83.3	81.0	84.1	90.0	92.2	94.4	79.0	87.2
	Method 2	89.1	94.8	96.7	95.1	88.3	84.9	89.0	93.5	96.1	93.8	85.8	91.5
	Method 3	91.0	95.4	97.5	94.2	92.3	86.9	91.2	95.6	97.3	96.4	86.6	93.0
	Method 4	92.6	96.6	96.8	95.5	93.5	89.3	92.2	96.5	98.2	96.9	91.5	94.5
	Method 5	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	94.0
	Method 6	92.9	96.7	97.1	96.4	94.7	90.4	92.9	97.2	98.5	98.1	92.3	95.2
	Method 7	94.5	96.9	97.4	96.1	95.4	94.0	94.8	97.6	98.5	97.7	94.9	96.2
	Method 8	**94.6**	**96.9**	**97.6**	**96.4**	**95.5**	**94.2**	**94.9**	**97.8**	**98.6**	**98.0**	**95.1**	**96.3**
In a coat#1-2	Method 1	61.4	75.4	80.7	77.3	72.1	70.1	71.5	73.5	73.5	68.4	50.0	70.4
	Method 2	70.7	85.5	86.9	83.3	77.1	72.5	76.9	82.2	83.8	80.2	66.5	78.7
	Method 3	76.0	87.6	89.8	85.0	81.2	75.7	81.0	84.5	85.4	82.2	68.1	81.5
	Method 4	76.6	90.0	90.3	87.1	84.5	79.0	84.1	87.0	87.3	84.4	69.5	83.6
	Method 5	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	77.4
	Method 6	80.0	91.8	93.0	87.8	86.5	82.9	85.2	90.0	90.8	89.3	78.4	86.9
	Method 7	82.2	93.0	95.2	91.6	87.1	83.4	87.2	90.1	92.4	88.2	75.8	87.8
	Method 8	**82.4**	**93.1**	**95.5**	**91.8**	**87.3**	**83.4**	**87.5**	**90.4**	**92.6**	**89.1**	**77.1**	**88.2**

5. Conclusion and future work

The proposed SEGait-ConViT exploits modified ConViT to yield the best performance. Since the SEGait-ConViT has the benefits of both ConViT and the added deeper layers with sparse edge-based feature extraction, it could achieve comparable results for person recognition using GEIs. Incorporating random convolutions into the self-attention process allows the model to capture global dependencies while still utilizing locality, a feature of convolutional networks. The proposed method's capacity to concentrate on local and global features is enhanced by this hybrid approach. The self-attention layers benefit greatly from the usage of random convolutions since they

provide an organized method for focusing on nearby areas in a random but significant way. This could be especially helpful in real-world situations where local areas of the image may be impacted by occlusions or distortions. Because the SEGait-ConViT framework concentrates on the most significant and discriminative aspects of the human body—edges that indicate motion and contour—it effectively captures crucial motion patterns in gait recognition. By extracting these robust, sparse features, the proposed model increases computational performance, decreases irrelevant information, and strengthens the model's resistance to problems like environmental change and occlusion. The SEGait-ConViT methodology achieved 98.8% and 66.5% accuracies on CASIA-B and Gait3D respectively. Future work will be to improve the accuracy by including multimodalities for feature extraction.

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